LECTURE 05

# Motion in a Straight Line

Chapter 2 — NCERT Physics Part I

Class 11 • CBSE + NEET • 2026–27

## Frame of Reference, Position, Distance and Displacement

2-hour session

**Observe → Define → Represent → Apply**

# What you will be able to do by the end

1. **Decide** whether an object is at rest or in motion relative to a stated observer
2. **Describe** the location of any object using an origin, direction and position coordinate
3. **Calculate** distance and displacement for straight-line and reversal paths
4. **Distinguish** scalar distance from vector displacement and state when they are equal
5. **Solve** NEET-pattern MCQs and CBSE 3-mark questions on this topic

**Mastery check: explain why distance can be non-zero when displacement is zero.**

MY NOTES

# Which instrument is telling the truth?

## You drive from home to school by a winding road.

**GPS / MAP**

### Reads: 10 km

Straight-line change in position from start to finish

**ODOMETER**

### Reads: 18 km

Total length of road actually travelled

**Neither is wrong. They measure different things — and naming those two things precisely is today's entire lecture.**

MY NOTES

# Recall — answer before we begin

1. What makes a physical quantity complete — not just a number?
2. What is the SI base unit of length?
3. Is speed a base quantity or a derived quantity? Justify in one sentence.
4. In one-dimensional motion, what does a negative sign on a value indicate?

MY NOTES

# Frame of reference

DEFINITION

**A frame of reference is the coordinate system relative to which position and motion are described.**

It sets the rules for measuring where and when things happen.

**ORIGIN (O)**

The reference point — assigned coordinate zero. Measurements start here.

**POSITIVE DIRECTION (+x)**

One direction is chosen as positive. Opposite is negative.

**CLOCK (t)**

Records when each observation is made. Position can change with time.

Key principle: shifting the origin changes every position coordinate but leaves the separation between any two points unchanged.

MY NOTES

# Position

**Position specifies the location of an object relative to the chosen origin.**

In one-dimensional motion, position is a single signed coordinate x.




**A : x = − 4 m**

**B : x = + 3 m**

**SI unit: metre (m) · Dimension: [L]**

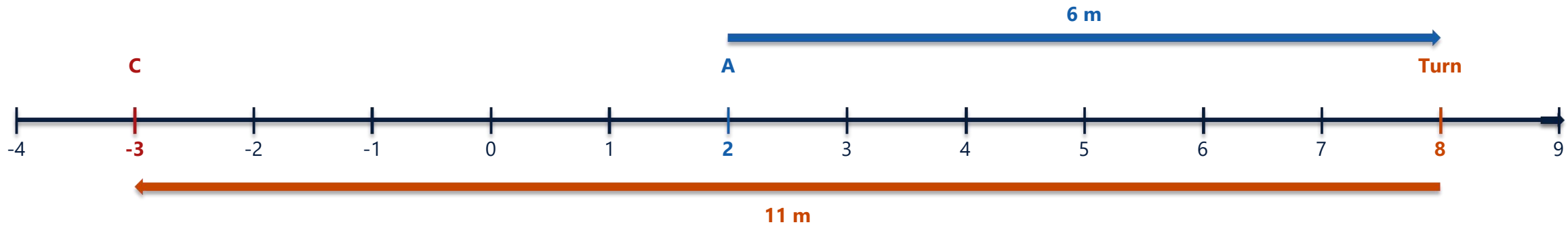
MY NOTES

# Distance

DISTANCE

**Total length of the actual path followed by an object.**

Scalar · always ≥ 0 · increases with every step · path-dependent · SI unit: metre




**Distance = 6 + 11 = 17 m**

Add every segment length — direction does not matter

MY NOTES

# Displacement

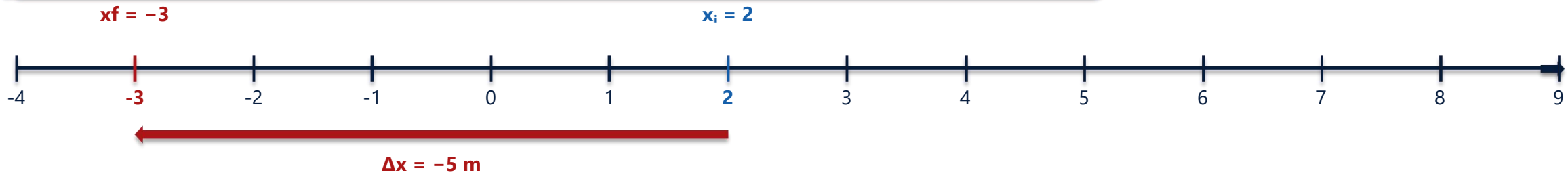
**DISPLACEMENT**

**Change in position from the initial point to the final point.**

Vector · can be positive, negative or zero · path-independent · SI unit: metre

$$\Delta x = x_{final} - x_{initial}$$

Subtract initial position from final — the sign gives the direction of net change




**Δx = (−3) − 2 = −5 m → 5 m in the negative direction**

MY NOTES

# Distance vs displacement — side by side

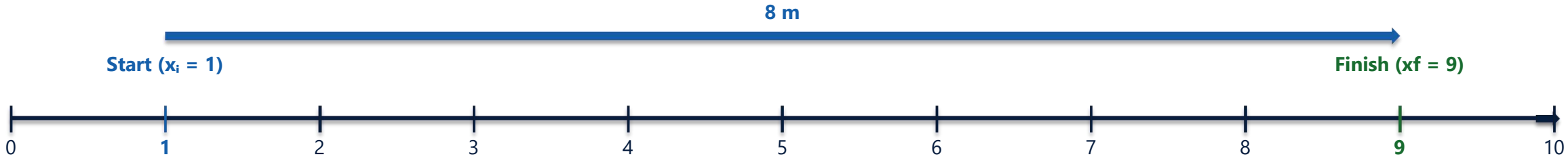
| Feature | Distance | Displacement |
|---|---|---|
| **Measures** | Total path length | Change in position |
| **Type** | Scalar | Vector |
| **Path** | Depends on path taken | Depends only on start and end |
| **Possible sign** | Zero or positive only | Positive, zero or negative |
| **SI unit** | metre (m) | metre (m) |

**Distance ≥ |Displacement|  Equality holds only when motion is along a straight line without any reversal.**

MY NOTES

# Straight-line path — when distance equals |displacement|

**An object moves from x = 1 m to x = 9 m in a straight line without reversing.**




**Distance = 8 m**

Straight path — one segment

**Δx = 9 − 1 = + 8 m**

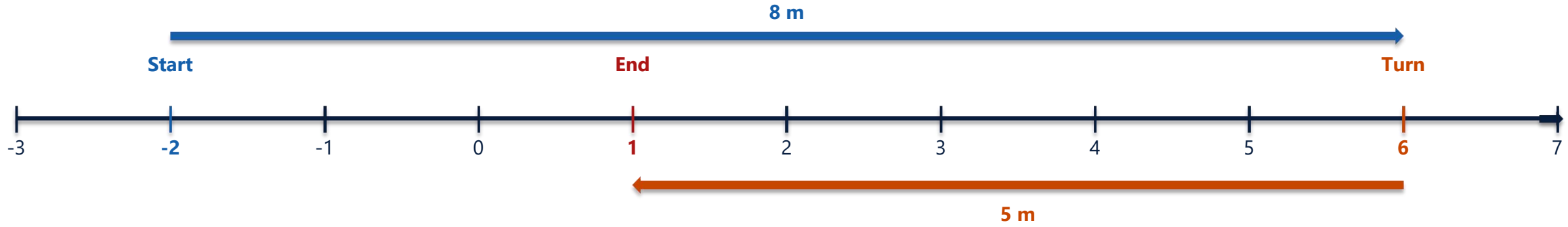
Direction: positive (+x)

**Distance = |Displacement| = 8 m ✓ This equality holds for any straight path with no reversal.**

MY NOTES

# Reversal of direction — when they differ

**Path: x = − 2 m → x = 6 m → x = 1 m**




**Distance = 8 + 5 = 13 m**

Sum both segment lengths — all positive

**Δx = 1 − (−2) = + 3 m**

xf − xi only — direction is positive

**NEET TRAP**

Do NOT compute distance by adding signed displacements: (+8) + (−5) = 3 m is the displacement, not the distance. Distance = 13 m; |Δx| = 3 m.

MY NOTES

# 5-MINUTE BREAK

Before you return: write one sentence distinguishing distance from displacement.

**Origin · Position x · Distance (path) · Displacement Δx = xf − xi**

# What does the sign of displacement mean?

## $\Delta x > 0$

**Net movement in the positive direction**

Finish is to the right of start

## $\Delta x < 0$

**Net movement in the negative direction**

Finish is to the left of start

## $\Delta x = 0$

**Start = Finish (position unchanged)**

At rest OR returned to start

**Zero displacement does NOT mean zero distance. A complete revolution returns $\Delta x = 0$ but distance = full circumference.**

MY NOTES

# Three traps that cost marks — NEET and CBSE

✗ Distance and displacement are always equal.

✓ Equal ONLY for a straight-line path with no reversal.

✗ Negative displacement means the object moved in the wrong direction.

✓ Negative means net movement opposite to the chosen positive axis — it is not an error.

✗ After one complete revolution, both distance and displacement are zero.

✓ Displacement = 0 (returned to start), but distance = circumference ≠ 0.

MY NOTES

# Solve — show full working

Show: path diagram → formula → substitution → unit → direction

1. An object starts at $x = -5$ m, moves to $x = +8$ m, then returns to $x = +2$ m. Find distance and displacement.

2. A ball rolls 12 m east then 5 m west. Find: (i) distance, (ii) displacement, (iii) is $|\text{displacement}| < \text{distance}$?

3. A runner completes exactly one lap of a 400 m circular track. Find distance and displacement.

MY NOTES

# Choose first — justify second

**Q1. Distance equals the magnitude of displacement when an object**

(A) moves in a curved path
(B) moves in a straight line without reversing direction
(C) completes a full circular revolution
(D) moves in a straight line and reverses once

**Q2. A particle completes one full revolution along a circular track. Its displacement is**

(A) equal to the circumference
(B) equal to the diameter
(C) zero
(D) equal to the radius

**Q3. If only the origin of a reference frame is shifted, which quantity changes for a fixed object?**

(A) Distance between two given points
(B) Position coordinate of each point
(C) Displacement between two given points
(D) Total path length

**Choose your answer first — then justify in one sentence.**

MY NOTES

# Answer individually — full working required

1. What minimum information is required before calling an object 'at rest'?

2. Write the position of an object located 7 m in the negative direction from the origin.

3. An object moves from $x = -2$ m to $x = 6$ m, then to $x = 1$ m. Calculate distance and displacement with full working.

4. A car completes exactly half a circular lap of radius 7 m. Find (i) distance and (ii) magnitude of displacement.

MY NOTES

# Lecture summary and next session

**FRAME OF REFERENCE**

Origin + direction + clock — sets the coordinate system for observation

**POSITION x**

Location of an object relative to the origin • Unit: m • Dim: [L]

**DISTANCE**

Total path length • Scalar • ≥ 0 • path-dependent

**DISPLACEMENT Δx**

xf − xi • Vector • Path-independent • Can be +, − or 0

**Next lecture: Average speed, average velocity and the position–time graph (x–t graph)**

MY NOTES